



22CH203 ENGINEERING CHEMISTRY-II

UNIT I & ELECTROCHEMISTRY

LESSON PLAN 1.2

Topic: Introduction to Electrochemistry

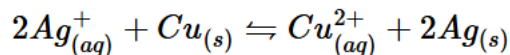
Reference: Chemistry for Engineering students by Lawrence S Brown and Thomas A Holme (Fourth Edition) Chapter 13

1. Voltaic or Galvanic Cells

Galvanic cells are electrochemical cells in which the electrons transferred due to redox reaction are converted to electricity.

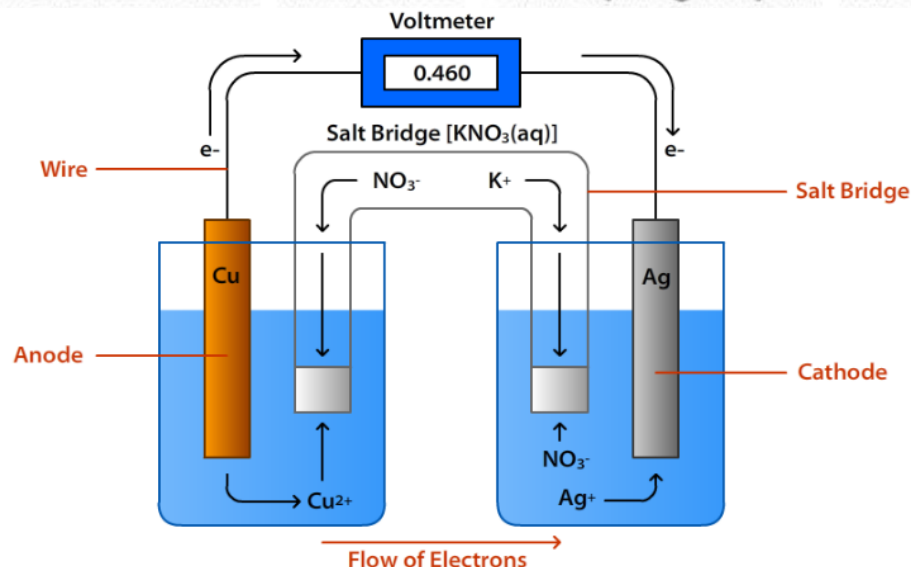
1.1 Example 1 Cu-Ag cell construction

Consider the reaction of a solid copper (Cu(s)) in a silver nitrate solution (AgNO₃(s)).



The AgNO₃(s) dissociates in water to produce Ag⁺(aq) ions and NO₃⁻(aq) ions. The NO₃⁻(aq) ions can be ignored since they are spectator ions and do not participate in the reaction. In this reaction, a copper electrode is placed into a solution containing silver ions. The Ag⁺(aq) will readily oxidize Cu(s) resulting in Cu²⁺(aq), while reducing itself to Ag(s).

This reaction releases energy. When the copper electrode solid is placed directly into a silver nitrate solution, however, the energy is lost as heat and cannot be used to do work. In order to harness this energy and use it do useful work, we must split the reaction into two separate half reactions; The oxidation and reduction reactions. A wire connects the two reactions and allows electrons to flow from one side to the other. In doing so, we have created a **Voltaic/ Galvanic Cell**.



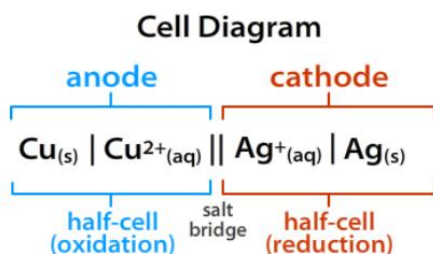
A Voltaic Cell (also known as a Galvanic Cell) is an electrochemical cell that uses spontaneous redox reactions to generate electricity. It consists of two separate **half-cells**. A half-cell is composed of an electrode (a strip of metal, M) within a solution containing M^{n+} ions in which M is any arbitrary metal. The two half cells are linked together by a wire running from one electrode to the other. A salt bridge also connects to the half cells. The functions of these parts are discussed below.

- **Half Cells** ∴ Half of the redox reaction occurs at each half cell. Therefore, we can say that in each half-cell a half-reaction is taking place. Here two halves of Cu and Ag are linked together with a wire and a salt bridge, an electrochemical cell is created
- **Electrodes** : An electrode is strip of metal on which the reaction takes place. In a voltaic cell, the oxidation and reduction of metals occurs at the electrodes. There are two electrodes in a voltaic cell, one in each half-cell. The **cathode** is where reduction takes place and oxidation takes place at the **anode**.
- Metal ion M^{n+} from the solution may collide with the electrode, gaining "n" electrons from it, and convert to metal atoms. This means that the ions are reduced. The cathode is where the reduction reaction takes place. This is where the metal electrode gains electrons. Referring back to the equation above, the cathode is the Ag(s) as it decreases in oxidation state from +1 to 0.
- Metal atom on the surface may lose "n" electrons to the electrode and enter the solution as the ion M^{n+} meaning that the metal atoms are oxidized. The anode is where the oxidation reaction takes place.

In other words, this is where the metal loses electrons. In the reaction above, the anode is the Cu(s) since it increases in oxidation state from 0 to +2.

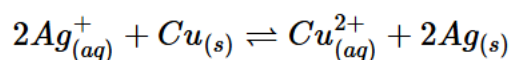
Flow of Electrons

Electrons always flow from the anode to the cathode or from the oxidation half cell to the reduction half cell. In terms of E°_{cell} of the half reactions, the electrons will flow from the more negative half reaction to the more positive half reaction. A cell diagram is a representation of an electrochemical cell. The figure below illustrates a cell diagram for the voltaic shown in Figure above.

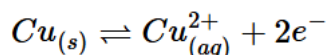
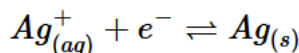


When drawing a cell diagram, we follow the following conventions. The anode is always placed on the left side, and the cathode is placed on the right side. The salt bridge is represented by double vertical lines (||). The difference in the phase of an element is represented by a single vertical line (|), while changes in oxidation states are represented by commas (,).

When asked to construct a cell diagram follow these simple instructions. Consider the following reaction:



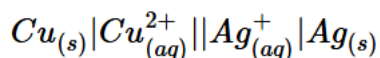
Step 1: Write the two half-reactions.



Step 2: Identify the cathode and anode.

- Cu(s) is losing electrons thus being oxidized; oxidation occurs at the anode. Anode (where oxidation occurs): $\text{Cu}_{(\text{s})} \rightleftharpoons \text{Cu}^{2+}_{(\text{aq})} + 2e^-$
- Ag⁺ is gaining electrons thus is being reduced; reduction happens at the cathode. Cathode (where reduction occurs): $\text{Ag}^+_{(\text{aq})} + e^- \rightleftharpoons \text{Ag}_{(\text{s})}$

Step 3: Construct the Cell Diagram.

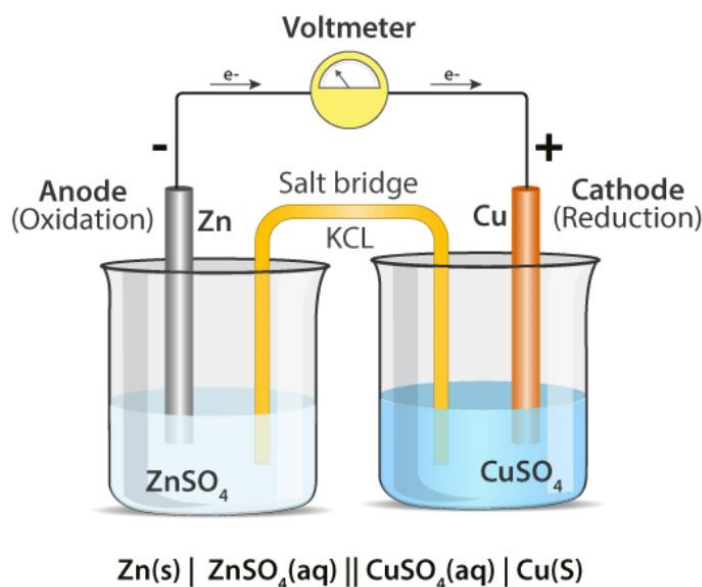


The anode always goes on the left and cathode on the right. Separate changes in phase by | and indicate the the salt bridge with ||. The lack of concentrations indicates solutions are under standard conditions (i.e., 1 M)

1.2 Example 2 : Construction of Daniel cell

A typical galvanic cell, it is designed to make use of the spontaneous redox reaction between zinc and cupric ion to produce an electric current.

This cell consists of a copper vessel. In which saturated $CuSO_4$ solution is filled which acts as depolarizer and $dil.H_2SO_4$ is filled which acts as an electrolyte. An amalgamated zinc rod is immersed in Zn_2SO_4 . In copper vessels there is a transparent layer all around just below the upper surface in which $CuSO_4$ crystals are kept in contact with $CuSO_4$ solution due to this the solution always remains saturated.



- The Daniell cell can be conventionally represented as



- In a Daniell cell electrons flow from zinc electrode to copper electrode through an external circuit, while metal ions form one half cell to the other through the salt bridge.

- Here current flows from copper electrode to zinc electrode that is cathode to anode via an external circuit.
- Daniell cell is a reversible cell while a voltaic cell may be reversible or irreversible.

1.2.1 Origin of Electrode potential

A metal (M) consists of metal ions (M^{n+}) with valence electrons. When the metal (M) is placed in a solution of its own salt, any one of the following reactions will occur.

- (i) Positive metal ions may pass into the solution.



- (ii) Positive metal ions from the solution may deposit over the metal



The above reactions indicate that the electrodes of a galvanic cell are at different potentials. So, it is necessary to know how potential arises in an electrode.

Illustration : Case 1: Zn electrode dipped in $ZnSO_4$ solution

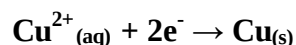
Zn goes into the solution as Zn^{2+} ions due to oxidation.



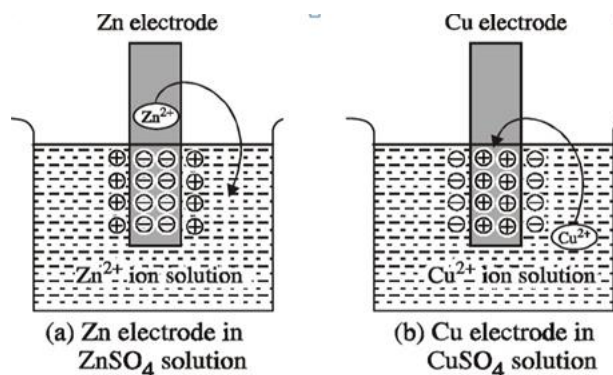
Now, the Zn electrode attains a negative charge, due to the accumulation of valence electrons on the metal. The negative charges developed on the electrode attract positive ions from solution.

Case 2: Cu electrode is dipped in $CuSO_4$ solution

Cu^{2+} ions from the solution deposit over the metal due to reduction

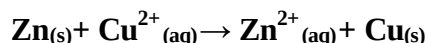


Now, the Cu electrode attains a positive charge, due to the accumulation of Cu^{2+} on the metal. The positive charges developed on the electrode attract negative ions from solution.





Net cell reaction (Redox reaction)



Thus, a sort of layer (positive or negative ions) is formed all around the metal. This layer is called **Helmholtz electrical double layer**. This layer prevents further passing of the positive ions from or to the metal. A difference of potential is consequently set up between the metal and the solution. At equilibrium the potential difference becomes a constant value, which is known as the electrode potential of the cell.

1.2.2 Factors affecting electrode potential

The rate of the above reactions depends on

The nature of the metal.

The temperature.

The concentration of metal ions in solution.

1.2.3 Single Electrode Potential (E)

It is the measure of the tendency of a metallic electrode to lose or gain electrons, when it is in contact with a solution of its own salt.

1.2.4 Standard Electrode Potential (E^0)

It is the measure of the tendency of a pure metallic electrode to lose or gain electrons, when it is in contact with a solution of its own salt of 1 molar concentration at 25°C.

1.2.5 Conditions for a cell to act as standard cell

The conditions for an electrochemical cell to act as a standard cell are

- *The e.m.f of the cell is reproductive.* The emf of external source is more than that of voltaic cells so that current may flow from external source into the voltaic cell and cell reaction can be reversed. If emf of voltaic cell is more than that of external source, current flows from voltaic cell into external source
- *The temperature-coefficient of e.m.f (change in e.m.f with temperature) should be very low.* Cell Reactions



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1.2.6 Salt bridge

It maintains electrical neutrality in two compartments by allowing movement of anions towards anodic compartment and cations towards cathodic compartment.

- It is a glass tube having potassium chloride or ammonium nitrate in a gelatin form.
- The gelatin allows ionic movement but prevents any kind of mixing.
- In the case of potassium chloride or ammonium nitrate the ionic mobilities of cations and anions are the same.

Functions of Salt bridge

- A salt bridge acts as an electrical contact between the two half cells.
- It prevents mechanical flow of solution, but it provides a free path for the migration of ions, to maintain an electric current through the electrolyte solution. It prevents the accumulation of charges.
- A salt bridge helps in maintaining the charge balance in the two half cells.
- A salt bridge minimizes or eliminates the liquid junction potential.

Discussion questions

1. Analyze the differences and similarities between the Daniell cell and modern batteries, such as lithium-ion batteries, in terms of materials, efficiency, and energy density.
2. Predict the effect of changing the concentration of electrolytes on the voltage and efficiency of a Daniell cell.
3. Discuss the environmental implications of using metals like zinc and copper in batteries.